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Assignment: Part 1

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Postgraduate Diploma in Data Analytics

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Part 1: Linear Regression

Plan of Analysis

Exploratory Data Analysis (EDA)

To understand the dataset completely and discover its patterns, trends, and correlations between variables, exploratory data analysis will be conducted.

- Make use of summary functions like .head(), .info(), and .describe() to load and look at the dataset

- Search for missing duplicate records and any missing values.
- Evaluate the way numerical variables which contains age, BMI and charges are allocated.
- Make use of graphs like boxplots, scatter plots and histograms to understand relationships.
- To determine correlations between variables, you have to use correlation matrix.

Feature Selection

The most important variables for determining costs of insurance will be discovered through feature selection.

- To identify variables that can be strongly associated with charges, conduct correlation analysis
- Make use of one-hot encoding to interpret categorical variables which include sex, smoker, and region into numerical form.
- Evaluation of the significance of features which depends on how they influence the target variable.
- Apply techniques which are like backward elimination if necessary to remove less significant variables

Model Training

To predict costs of insurance, a linear regression model will be implemented.

- Apply an 80 over 20 ratio to breakdown the dataset into training and testing sets.
- Select the training data set to train your model.
- Starting off, use the Linear Regression algorithm's default parameters.
- To understand the relationship between independent and target variables, fit the model.

Model Evaluation

To figure out the accuracy and effectiveness of the model, its performance will be evaluated using appropriate metrics.

- R2 Score is used to evaluate how well the model will have to adjusts for the data's risk.
- To get the average estimate error, Mean Absolute Error (MAE) have to be applied.
- Root Mean Squared Error (RMSE) to serve as understandable error measurement.

Report Writing

The results of the analysis will be provided in a thorough report.

Which will include:

- Conclusions and recommendations
- The dataset's description
- Preprocessing and data cleaning procedures
- EDA insights and visualisation
- The process of training models
- Results and Interpretation of assessment
- Conclusions and recommendations

Introduction

To maintain fair and long-term pricing, medical aid programs seek to precisely determine their client's healthcare costs. Lifestyle and demographic factors such as age, body mass index (BMI), smoking habits, and location are frequently an impact on these expenses.
The aim of this research is to develop a Linear Regression model that can evaluate medical insurance costs depending on client attributes. This project shows how predictive models will assist medical aid providers anticipate expenses more accurately through the use of data analytics techniques on the Insurance dataset.

Data Description

The insurance dataset , which was captured through Kaggle was used in this investigation. It contains 7 variables and 1338 records that explain the medical insurance charges and characteristics of the customers.

The dataset contains categorical variables, which are sex, smoker, and region and also numerical variables which includes age, BMI, number of children, and charges.

The dataset is suitable for analysis and modelling given that it is complete and does not have any missing values or duplicate records.

Variables	**Description**	**Types**
age	Age of the customer	Numerical
sex	Gender of the customer	Categorical
bmi	Body Mass Index	Numerical
children	Number of dependents	Numerical
smoker	Smoking status	Categorical
region	Customer's residential region	Categorical
charges	Costs of Insurance	Numerical

Data Cleaning

The dataset was examined to make sure its quality and suitability for modelling just before conducting the analysis.

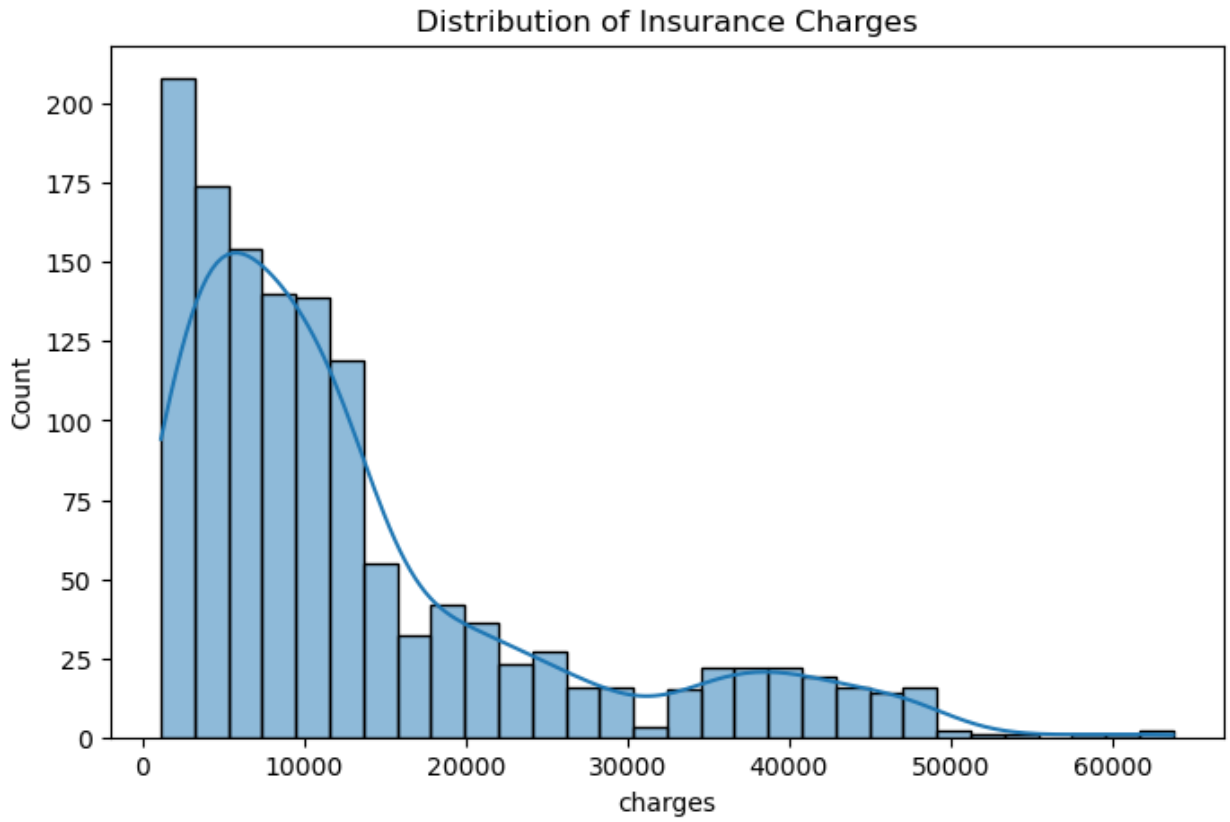
- Looking for any missing values
- Look for records that were duplicates
- Confirmed data types

There were no duplicate entries or any missing values in the dataset, which indicates that no major cleanup was required.

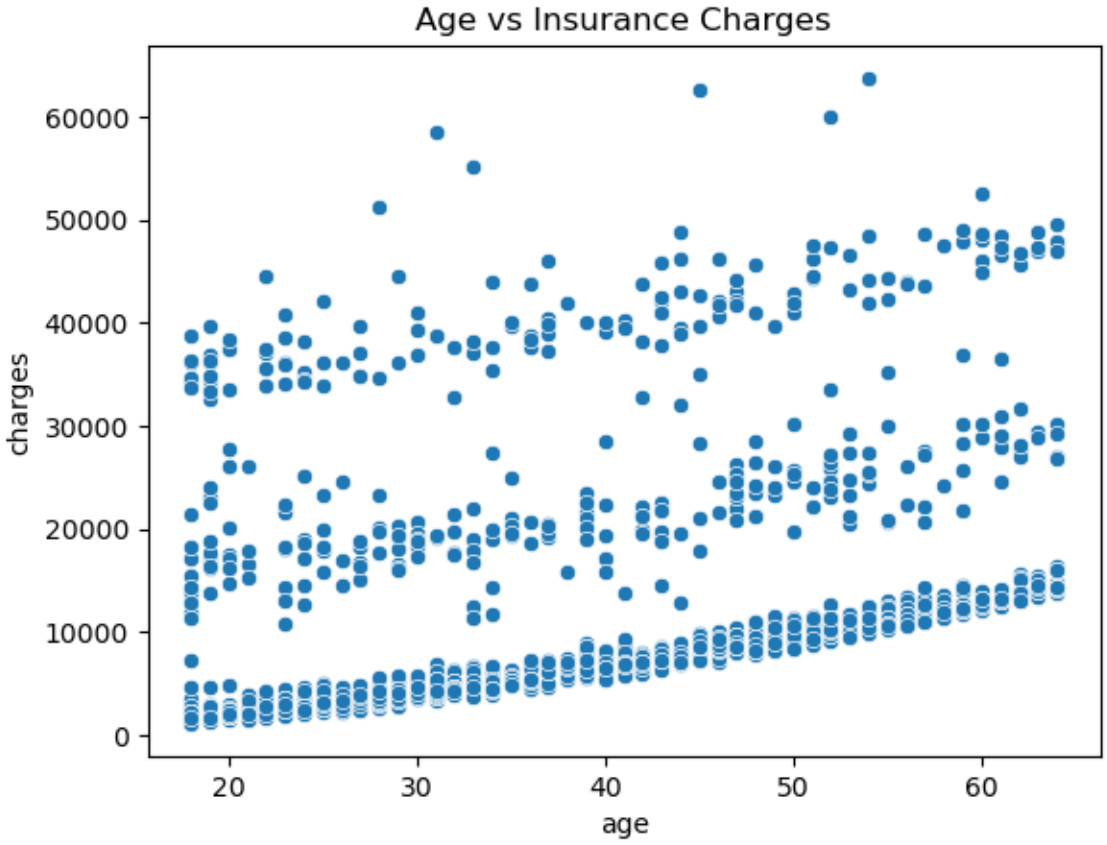
Instead, one-hot encoding has been implemented to change categorical information such as sex, smoker, and region into numerical format. Given that machine learning algorithms demand numerical input data, this step was required.

Exploratory Data Analysis (EDA)

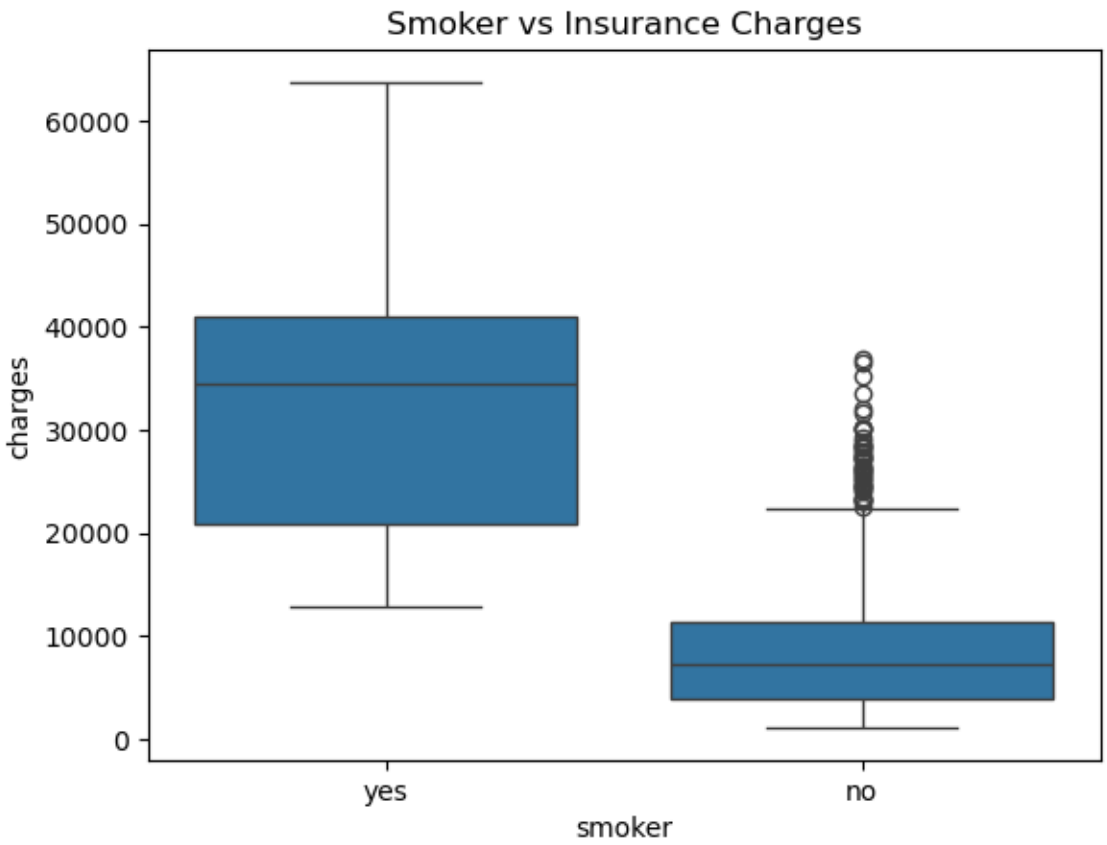
To determine trends, connections, and patterns in the dataset, exploration data analysis was implemented.



The majority of people have moderate healthcare prices, even though a smaller percentage have very high charges, according to a histogram of the charges variable.



Age and charges revealed a positive correlation in scatter plot, showing that older people usually have higher medical costs.



Insurance premiums for smokers and non-smokers differed considerably, with smokers paying much higher, according to a boxplot. This suggests that smoking has a major impact on costs of healthcare.

Furthermore, study of correlations revealed a strong relationship between insurance costs and factors which include age, BMI, and smoking status.

Model Training

80 over 20 ratios have been used to break down the dataset into training and testing sets, using 80% of the data getting used to train the model and 20% which was used to evaluate its performance.

The Scikit-learn library's Linear Regression model was implemented. To make sure that it teaches the model the relationship between the independent variables and the target variable, it was adjusted to the training data.

The model was used to estimate the cost of the insurance on the test dataset once it is fully trained.

Model Evaluation

The model's performance was evaluated using several metrics:

Metric	Value
R^2 Score	(0.7835929767120724)
MAE	(4181.1944737536505)
MSE	(33596915.851361446)
RMSE	(5796.29)

The model's mobility to explain the differences in rates of the insurance is demonstrated by the R2 value. A better fit is shown by a higher value.

The average gap between estimated and actual data is calculated by the Mean Absolute Error (MAE)

The amount of prediction mistakes can be determined from the Mean Squared Error (MSE) and Root Mean Squared Error (RMSE)

In general, based on the selected features, the model presents a respectable capacity to predict medical insurance expenses.

Model Improvement

The model performed adequately using default parameters. However, several improvements could be made:

- Implementing feature scaling
- Minimising less important variables
- Evaluating more advanced models such as Ridge or Lasso regression
- Carrying out hyperparameter change

Conclusion

Using the data of the customer, this study successfully developed a Linear Regression model to estimate medical insurance costs. The outcomes suggested that variables which includes age , BMI, and smoking status have a big effect on the costs of insurance.

The model indicates the way machine learning and data analytics could assist medical aid providers to evaluate expenses and provide data-driven decisions.

To further improve accuracy in forecasting, future studies could make research more complex modeling techniques.

References

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https://www.kaggle.com/datasets/mirichoi0218/insurance
[Accessed March 2026].

Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*.